Uncertainty across volatility regimes: Evidence from an updated U.S. SVAR

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Abstract

This paper investigates whether macroeconomic and financial uncertainty are exogenous drivers of U.S. business-cycle fluctuations or mainly endogenous responses to real activity, addressing the identification problem posed by endogeneity and reverse causality. Identification is achieved within a nonrecursive structural VAR framework that exploits breaks in the unconditional covariance matrix of reduced-form innovations thanks to the identification through heteroskedasticity method.

The analysis explicitly extends the regime-based nonrecursive SVAR framework developed by Angelini, Bacchiocchi, Caggiano, and Fanelli (2019) to a longer and updated monthly dataset up to June 2025, covering major post-2015 developments including the COVID-19 pandemic and the subsequent high-volatility period. Uncertainty measures are taken from Sidney Ludvigson's data repository, while real activity is proxied by industrial production from FRED.

Results confirm that macroeconomic uncertainty behaves as an exogenous source of business-cycle fluctuations, producing contractionary effects on real activity across all regimes, with markedly stronger and more persistent responses during high-volatility regimes. Financial uncertainty affects real activity mainly indirectly through its impact on macroeconomic uncertainty, with amplification becoming more relevant from the mid-1980s onward. Overall, the findings highlight that the regime-dependent nature of uncertainty shocks remains empirically relevant in the most recent macroeconomic environment.

1 Introduction

The role of uncertainty as a source of macroeconomic fluctuations has become a central theme in modern macroeconomics, particularly in the aftermath of major episodes of economic and financial distress. A large empirical literature documents that increases in uncertainty are systematically associated with declines in real economic activity, while uncertainty itself tends to rise sharply during recessions and periods of financial turmoil. These stylized facts raise a fundamental question that remains unresolved: does uncertainty represent an exogenous

driver of the business cycle, or is it mainly an endogenous response to adverse macroeconomic conditions?

Answering this question is empirically challenging because uncertainty and real activity are likely to be jointly determined. Most commonly used measures of macroeconomic and financial uncertainty are constructed from large information sets of macroeconomic and financial variables that strongly co-move with the business cycle. As emphasized by Ludvigson, Ma and Ng (2018), this feature naturally amplifies concerns about endogeneity and reverse causality, making it difficult to interpret estimated uncertainty shocks as structural disturbances. Standard recursive SVARs typically address this issue by imposing timing restrictions that rule out contemporaneous feedback from real activity to uncertainty by construction. While convenient, such assumptions are difficult to justify in monthly data and may lead to distorted inference regarding the causal role of uncertainty.

Recent contributions have therefore proposed alternative identification strategies designed to relax recursive assumptions and explicitly address reverse causality. A prominent line of research exploits changes in volatility across macroeconomic regimes as a source of identifying information, following the identification-through-heteroskedasticity approach pioneered by Rigobon (2003) and Lanne and Lütkepohl (2008). The key insight is that breaks in the covariance matrix of VAR innovations generate additional moment conditions that can be used to identify structural shocks without relying on exclusion restrictions. This approach is particularly appealing in the U.S. context, where the postwar period is characterized by well-documented shifts in volatility associated with major historical episodes such as the Great Inflation, the Great Moderation, and the Global Financial Crisis.

Building on this literature, Angelini, Bacchiocchi, Caggiano and Fanelli (2019) develop a nonrecursive SVAR framework that exploits breaks in the unconditional covariance matrix of reduced-form innovations while allowing the contemporaneous transmission mechanism itself to vary across volatility regimes. This methodology jointly addresses two key issues that are often treated separately: the exogeneity versus endogeneity of uncertainty, and the possibility that the effects of uncertainty shocks depend on the prevailing macroeconomic environment. Their empirical findings suggest that macroeconomic uncertainty behaves as an exogenous driver of business cycle fluctuations, while financial uncertainty affects real activity mainly indirectly through macroeconomic uncertainty, with effects that amplify during periods of heightened volatility.

The present paper extends this empirical strategy along an important dimension. Specifically, it reassesses the causal role of macroeconomic and financial uncertainty using a substantially longer and more recent dataset that extends to June 2025. This extension is particularly informative because it encompasses the post-2015 period, the COVID-19 pandemic, and the subsequent phase of elevated macroeconomic and financial volatility. These events represent an unprecedented stress test for the uncertainty–business cycle relationship and provide a natural setting in which to evaluate the stability of earlier conclusions in an updated macroeconomic

environment.

The empirical analysis focuses on U.S. monthly data and considers a small-scale VAR including macroeconomic uncertainty, real economic activity, and financial uncertainty. Uncertainty measures are taken from the datasets made available by Sidney Ludvigson, building on Jurado, Ludvigson and Ng (2015) and Ludvigson, Ma and Ng (2018), while real activity is proxied by industrial production from the FRED database. The identification strategy exploits breaks in the unconditional covariance matrix of VAR innovations and allows for contemporaneous interactions among the variables, thereby permitting reverse causality to be tested rather than ruled out by assumption. Relative to standard heteroskedastic SVARs with regime-invariant structural parameters, the adopted framework allows both volatility and the contemporaneous transmission mechanism to differ across regimes, a feature that is especially relevant in the presence of large structural changes.

The main findings of the paper can be summarized as follows. First, macroeconomic uncertainty is consistently identified as an exogenous driver of business cycle fluctuations across all volatility regimes. Second, the contractionary effects of macroeconomic uncertainty shocks on real activity are substantially stronger and more persistent during high-volatility regimes, with particularly pronounced effects in the post-2008 period and in the post-pandemic environment. Third, financial uncertainty affects real activity mainly indirectly through its impact on macroeconomic uncertainty, with amplification effects becoming more relevant from the mid-1980s onward.

The remainder of the paper is organized as follows. Section 2 discusses the reverse-causality problem and motivates the identification strategy. Section 3 presents the econometric framework and identification through changes in volatility. Section 4 reports the empirical results. Section 5 concludes.

2 The Reverse Causality Issue

Understanding whether uncertainty represents a causal force behind business-cycle fluctuations or, instead, an endogenous outcome of macroeconomic dynamics is a central issue in modern macroeconomics. A large empirical literature documents that sharp increases in uncertainty are systematically associated with contractions in real economic activity, weaker output growth, and declines in employment and investment. Jurado, Ludvigson, and Ng (2015) show that broad-based measures of macroeconomic uncertainty are strongly negatively correlated with real activity, while Ludvigson, Ma, and Ng (2018) further emphasize the pervasive macroeconomic effects of uncertainty shocks across a wide range of aggregates.

At the same time, uncertainty displays a pronounced countercyclical pattern, rising during recessions and episodes of financial stress. This evidence suggests that uncertainty may not only act as a source of macroeconomic fluctuations, but may also respond endogenously to adverse economic conditions. Bachmann and Moscarini (2012) highlight feedback mechanisms between

labor-market conditions and uncertainty, while Carriero, Clark, and Marcellino (2018) show that measures of uncertainty tend to increase following negative macroeconomic and financial shocks. Taken together, these findings point to potentially strong two-way interactions between uncertainty and the business cycle.

These regularities give rise to a fundamental identification challenge. It remains unclear whether macroeconomic and financial uncertainty should be interpreted primarily as exogenous drivers of economic fluctuations or whether they mainly reflect endogenous responses to other structural disturbances affecting output, demand, supply, or financial conditions. Disentangling these interpretations is crucial for both macroeconomic theory and policy analysis, as it directly affects how uncertainty shocks are modeled and how their macroeconomic consequences are evaluated.

From a conceptual standpoint, two broad views have emerged. In the first, uncertainty is treated as an exogenous structural shock: increases in uncertainty depress economic activity through real-options and "wait-and-see" effects, precautionary behaviour, and amplification via financial frictions. This view is central to analyses such as Christiano, Motto, and Rostagno (2014) and Basu and Bundick (2017). In the second view, uncertainty is largely endogenous, responding to changes in fundamentals and to adverse macroeconomic or financial conditions rather than originating as an independent disturbance, as argued by Bachmann and Moscarini (2012) and Gourio (2014). The distinction matters for policy: if uncertainty is exogenous, it is a fundamental shock; if it is mostly endogenous, its countercyclical behavior mainly reflects the propagation of other shocks.

Empirically, separating exogeneity from endogeneity is difficult. Measures of uncertainty are often constructed from information sets that co-move with the business cycle, making contemporaneous feedback effects plausible. Recursive SVARs identified through Cholesky decompositions address the problem only by assumption, imposing timing restrictions that rule out within-period feedback from real activity to uncertainty by construction. In monthly data, such restrictions are hard to defend and may distort structural inference when reverse causality is relevant.

The identification problem is further complicated by structural instability and pronounced changes in volatility in U.S. macro-financial data. The postwar period features distinct volatility regimes commonly associated with the Great Inflation, the Great Moderation, and the post 2007/08 period. Rather than treating these shifts solely as a nuisance, the change-in-volatility literature shows that breaks in second moments can be exploited as identifying information: shifts in the reduced-form covariance matrix generate additional moment conditions that help identify structural shocks without relying on contemporaneous exclusion restrictions.

This paper addresses the reverse-causality issue by adopting a nonrecursive SVAR identified through breaks in the reduced-form innovation covariance matrix (identification through heteroskedasticity). The empirical objective is to assess whether macroeconomic and financial uncertainty act as exogenous drivers of the business cycle or mainly reflect endogenous responses

to other shocks. The analysis uses updated uncertainty measures from Sidney Ludvigson's data repository together with macroeconomic series from FRED, allowing the conclusions to be evaluated in a sample that includes the post-2019 period.

In sum, two requirements guide the empirical design. First, an identification strategy is needed that allows for contemporaneous interactions between uncertainty and real activity, so that reverse causality can be tested rather than assumed away. Second, the analysis should accommodate volatility regimes and allow shock transmission to differ across them. The change-in-volatility nonrecursive SVAR framework addresses both dimensions in a unified way.

3 Methodology

This section presents the econometric framework and the identification strategy used to analyze the dynamic causal interaction between uncertainty and real economic activity. The key methodological choice is to move beyond standard recursive Cholesky SVAR identification, which is potentially ill-suited in the presence of contemporaneous feedback and reverse causality, and to adopt a nonrecursive identification-through-heteroskedasticity approach based on changes in volatility across regimes. The framework builds on Bacchiocchi and Fanelli (2015), its extension in Bacchiocchi, Castelnuovo and Fanelli (2018), and its application to uncertainty shocks in Angelini, Bacchiocchi, Caggiano and Fanelli (2019).

3.1 Reduced-form VAR

Let

$$X_t = \left(U_t^M,\ Y_t,\ U_t^F\right)', \tag{1}$$

where U_t^M denotes macroeconomic uncertainty, Y_t is a measure of real economic activity, and U_t^F is financial uncertainty. The reduced-form VAR(p) is specified as

$$X_t = \Pi(t)W_t + \eta_t, \qquad t = 1, \ldots, T, \tag{2}$$

where $W_t = (X'_{t-1}, \ldots, X'_{t-p}, 1)'$, $\Pi(t)$ collects possibly regime-dependent reduced-form coefficients, and η_t is a zero-mean innovation with unconditional covariance matrix

$$\Sigma_\eta(t) = \mathbb{E}(\eta_t \eta_t'). \tag{3}$$

A key empirical feature of the data is that $\Sigma_\eta(t)$ is not constant over time. Instead, the reduced-form innovations display distinct volatility regimes commonly associated with the Great Inflation (GI), the Great Moderation (GM), and the Great Recession plus Slow Recovery (GR+SR). Rather than treating these shifts solely as a nuisance, the framework exploits them as a source of identifying information.

3.2 Structural form and the identification problem

Reduced-form innovations are linked to structural shocks through

$$\eta_t = B(t)e_t, \qquad \mathbb{E}(e_t e_t') = I_3, \tag{4}$$

where $e_t = (e_t^M, e_t^Y, e_t^F)'$ represent macroeconomic uncertainty, real-activity, and financial uncertainty shocks, respectively. The matrix $B(t)$ collects contemporaneous on-impact effects.

Without additional restrictions, $B(t)$ is not identified from $\Sigma_\eta(t)$ because

$$\Sigma_\eta(t) = B(t)B(t)'. \tag{5}$$

In a homoskedastic environment, this mapping does not uniquely pin down the structural parameters and a common solution is to impose a recursive structure on $B(t)$, discussed next.

3.3 Recursive identification and its limitations

A common approach in applied structural VAR analysis is to impose a recursive (Cholesky) identification scheme by restricting the contemporaneous transmission matrix to be triangular. One convenient representation of such a model is

$$A_0 X_t = A_1 X_{t-1} + \cdots + A_p X_{t-p} + u_t, \qquad \mathbb{E}(u_t u_t') = I_3, \tag{6}$$

where A_0 is nonsingular. Premultiplying by A_0^{-1} yields the reduced form in (2), with $\eta_t = A_0^{-1} u_t$ and $B = A_0^{-1}$. Identification is achieved by imposing a triangular structure on A_0, typically lower triangular with positive diagonal elements, which implies a recursive causal ordering among the variables.

In the context of uncertainty and real activity, such an ordering has strong implications. Depending on the chosen ordering, recursive identification typically implies either that real activity does not contemporaneously affect uncertainty or that uncertainty does not contemporaneously affect real activity. In both cases, one direction of contemporaneous causality is ruled out by assumption rather than tested empirically. This feature is particularly problematic in monthly data, where it is plausible that uncertainty and real activity respond to common information sets, news, and shocks essentially within the same period.

The difficulty is amplified by the nature of commonly used uncertainty measures. Macroeconomic and financial uncertainty indices are typically constructed from large panels of macroeconomic and financial variables that are themselves closely linked to real economic activity. As a result, contemporaneous feedback from real activity to measured uncertainty is not only possible but likely. Imposing exclusion restrictions that preclude such feedback therefore requires strong timing assumptions that are difficult to justify empirically.

Moreover, recursive identification is inherently tied to a fixed causal ordering that is assumed

to be invariant over time. This feature can be particularly restrictive in environments characterized by structural change and regime shifts. If the contemporaneous relationships among uncertainty, real activity, and financial conditions evolve across macro-financial regimes, a single recursive ordering may provide a poor approximation to the true data-generating process. In such cases, recursive SVARs may deliver distorted estimates of structural shocks and misleading impulse responses.

These concerns are reinforced by evidence showing that recursive and nonrecursive identification schemes can lead to markedly different conclusions once breaks and volatility regimes are explicitly taken into account. In settings with pronounced heteroskedasticity and structural instability, recursive identification may confound changes in shock variances with changes in transmission mechanisms, obscuring the true causal role of uncertainty.

Taken together, these considerations suggest that recursive identification is ill-suited for addressing the reverse-causality problem at the core of the uncertainty–business cycle debate. This motivates the adoption of a nonrecursive identification strategy that allows for contemporaneous interactions and exploits changes in volatility as a source of identifying information, rather than relying on timing restrictions that may be invalid in the presence of endogeneity and regime shifts.

3.4 Identification through changes in volatility

Assume that the unconditional covariance matrix of reduced-form innovations exhibits $R = 3$ regimes separated by two break dates T_{B1} and T_{B2}:

$$\Sigma_\eta(t) = \begin{cases} \Sigma_{\eta,1}, & t \leq T_{B1}, \\ \Sigma_{\eta,2}, & T_{B1} < t \leq T_{B2}, \\ \Sigma_{\eta,3}, & t > T_{B2}. \end{cases} \tag{7}$$

A prominent approach, associated with Lanne and Lutkepohl (2008), building on Rigobon (2003), assumes a time-invariant impact matrix B and attributes the regime shifts solely to changes in the variances of the structural shocks. In the three-regime case, this implies a structure of the form:

$$\Sigma_{\eta,1} = BB', \qquad \Sigma_{\eta,2} = B\Lambda_2 B', \qquad \Sigma_{\eta,3} = B\Lambda_3 B', \tag{8}$$

where Λ_2 and Λ_3 are diagonal matrices with positive diagonal elements. Under suitable distinctness conditions on the diagonal elements, B can be identified up to sign permutations. A crucial implication is that dynamic causal effects are regime-invariant in shape: regimes differ mainly by scaling.

Following Bacchiocchi and Fanelli (2015) we relax the restrictive assumption embedded in

(8) and allow the contemporaneous transmission to change across regimes:

$$\eta_t = \begin{cases} B\, e_t, & t \le T_{B1}, \\ (B + Q_2)e_t, & T_{B1} < t \le T_{B2}, \\ (B + Q_2 + Q_3)e_t, & t > T_{B2}, \end{cases} \tag{9}$$

so that the regime-specific covariance matrices satisfy

$$\begin{aligned} \Sigma_{\eta,1} &= BB', \\ \Sigma_{\eta,2} &= (B + Q_2)(B + Q_2)', \\ \Sigma_{\eta,3} &= (B + Q_2 + Q_3)(B + Q_2 + Q_3)'. \end{aligned} \tag{10}$$

This generalization is central: it allows the on-impact causal links among uncertainty and real activity to vary across macro-financial environments. Importantly, it also delivers regime-dependent IRFs not only in magnitude but potentially in sign, persistence, and timing, consistent with the empirical motivation in Angelini, Bacchiocchi, Caggiano, Fanelli (2019).

3.5 Moment conditions, order and rank conditions

Let θ collect the free structural parameters in (B, Q_2, Q_3) once identifying restrictions are imposed. Define the vector of distinct reduced-form second moments across regimes as

$$\sigma^+ = \begin{pmatrix} vech(\Sigma_{\eta,1}) \\ vech(\Sigma_{\eta,2}) \\ vech(\Sigma_{\eta,3}) \end{pmatrix}, \tag{11}$$

where $vech(\cdot)$ stacks the lower-triangular elements of a symmetric matrix. The structural model implies a differentiable mapping

$$\sigma^+ = g(\theta), \tag{12}$$

induced by the moment restrictions (10). Identification requires:

(i) Order condition. With $n = 3$ variables and $R = 3$ regimes, the number of distinct reduced-form covariance elements is

$$r = R \cdot \frac{n(n+1)}{2} = 3 \cdot \frac{3 \cdot 4}{2} = 18. \tag{13}$$

Hence, the necessary condition is $\dim(\theta) \le r$. Since (B, Q_2, Q_3) contains $3n^2 = 27$ elements absent restrictions, at least $27 - 18 = 9$ restrictions are needed to meet the order condition.

(ii) Rank condition. The necessary and sufficient condition for local identification is that the Jacobian

$$J(\theta) := \frac{\partial g(\theta)}{\partial \theta'} \tag{14}$$

has full column rank in a neighborhood of the true parameter θ_0. In practice, $J(\theta)$ can be evaluated analytically or numerically at the estimated parameters, providing an ex-post verification that the imposed restrictions are sufficient to uniquely pin down the structural parameters up to sign normalizations.

An important advantage of this framework is that it can accommodate both exactly identified and overidentified structures. Overidentifying restrictions implied by special cases (e.g., the time-invariant impact matrix in (8)) are testable via likelihood-based or Wald statistics.

3.6 Nonrecursive structure, exogeneity, and impulse responses

Because the contemporaneous impact matrix is not restricted to be triangular, the model allows for instantaneous feedback between uncertainty and real economic activity. This feature is central in addressing the reverse-causality issue, as it permits uncertainty to affect real activity on impact while simultaneously allowing real activity to influence uncertainty within the same period. Hypotheses concerning exogeneity are formulated as linear restrictions on the elements of the regime-specific impact matrices and can be tested using standard likelihood-based inference within the change-in-volatility identification framework.

Regime-dependent impulse response functions are obtained by combining the estimated contemporaneous impact matrices with the VAR dynamics. Let A_r denote the companion-form matrix associated with the autoregressive dynamics in volatility regime r, and let B_r denote the corresponding contemporaneous impact matrix, with $B_1 = B$, $B_2 = B + Q_2$, and $B_3 = B + Q_2 + Q_3$. The impulse response of X_{t+h} to a one-standard-deviation structural shock $e_{j,t}$ in regime r is given by

$$IRF_j^{(r)}(h) = R'(A_r)^h R\, b_{j,r}, \qquad h = 0, 1, \ldots, h_{\max}, \tag{15}$$

where $b_{j,r}$ denotes the j-th column of B_r and R is a selection matrix that maps the companion-form state vector back to the observable variables.

Even when the autoregressive dynamics are restricted to be identical across regimes, impulse responses differ across volatility regimes because the contemporaneous transmission mechanism embodied in B_r varies with r. As a result, regime dependence in the impulse responses is not limited to scale effects but may also involve differences in sign, timing, persistence, and overall dynamic shape. This feature represents a key departure from standard heteroskedastic SVARs with time-invariant impact matrices, where regime shifts affect impulse responses only through changes in shock variances.

Allowing both contemporaneous interactions and regime-dependent transmission provides

a flexible framework for assessing state-dependent propagation mechanisms. In the context of uncertainty shocks, this flexibility is particularly important, as periods of heightened macro-financial stress may alter not only the size of uncertainty shocks but also the way they propagate through the economy. The regime-specific impulse responses derived from this model therefore offer a natural tool for evaluating whether uncertainty behaves as an exogenous driver of economic fluctuations and whether its effects depend systematically on the prevailing volatility environment.

The next section turns to estimation and to the empirical implications of the identified regime-dependent SVAR.

4 Empirical Results

This section presents the empirical evidence obtained from the nonrecursive SVAR identified through changes in volatility. The focus is on the dynamic interaction between macroeconomic uncertainty, real economic activity, and financial uncertainty across volatility regimes.

Real activity is measured by the monthly growth rate of industrial production, computed as the first difference of the log level of the industrial production index. This transformation is standard in monthly VAR applications and provides a stationary proxy for real economic activity. Uncertainty measures are used in levels and are considered at both short and long horizons, corresponding to one-month-ahead ($f = 1$) and one-year-ahead ($f = 12$) uncertainty. This distinction reflects the forward-looking nature of uncertainty and allows us to assess whether short-lived and more persistent uncertainty components propagate differently over the business cycle. The VAR includes a constant and is estimated with $p = 4$ lags, a conventional choice for monthly data that balances dynamic flexibility and parsimony and is consistent with lag structures commonly adopted in related small-scale systems.

Identification relies on shifts in the unconditional covariance matrix of reduced-form innovations. Volatility regimes are selected based on rolling and recursive evidence on second moments, complemented by formal Chow-type likelihood-ratio tests for breaks in the covariance structure. Two break dates are used to partition the sample into three regimes corresponding to the Great Inflation (GI), the Great Moderation (GM), and the Great Recession plus Slow Recovery (GR+SR). This regime classification is in line with well-documented changes in U.S. macroeconomic volatility and closely follows the partition adopted in the regime-based SVAR literature.

Impulse response functions are computed over a horizon of $h_{\max} = 60$ months and are normalized so that each structural shock generates a positive contemporaneous response in its own variable. Inference is conducted using 90% bootstrap confidence bands constructed via a regime-specific residual bootstrap that preserves both the estimated break structure and the heteroskedastic nature of the VAR innovations.

4.1 Bootstrap inference

The confidence bands reported in Figures 3–5 are constructed using a residual-based bootstrap that resamples reduced-form VAR innovations within each volatility regime. Specifically, residuals are drawn with replacement separately for each regime, and artificial samples are generated by iterating the estimated VAR while preserving the original break dates and the regime-specific covariance structure. This procedure ensures that the heteroskedasticity patterns underlying identification are maintained in the bootstrap data.

Preserving regime-specific second moments is crucial in a framework where identification relies on changes in the unconditional covariance matrix of reduced-form innovations. Standard i.i.d. bootstraps, which ignore volatility regimes, or asymptotic inference based on the delta method, may fail to adequately capture the sampling uncertainty associated with structural parameters when heteroskedasticity and structural breaks are present. In contrast, the regime-specific bootstrap explicitly accounts for both parameter uncertainty and the nonstandard features induced by volatility shifts.

From a theoretical perspective, this bootstrap approach provides a consistent approximation to the finite-sample distribution of impulse response functions in heteroskedastic SVARs with structural breaks, as it jointly resamples the reduced-form innovations and propagates uncertainty through the estimated structural mapping. As a result, the resulting confidence bands appropriately reflect the combined uncertainty stemming from estimation of the VAR dynamics, the regime-dependent impact matrices, and the break structure. For these reasons, the regime-specific residual bootstrap represents a natural and reliable inference procedure in the present change-in-volatility identification framework.

4.2 Volatility regimes and covariance breaks

Before turning to structural dynamics, it is useful to document the empirical evidence on volatility shifts that underpins identification. Table 1 reports maximum-likelihood estimates of the covariance and implied correlation matrices of reduced-form VAR innovations for the full sample and for each volatility regime. A Chow-type likelihood-ratio test strongly rejects the null hypothesis of no breaks in the innovation covariance matrix, providing formal support for the existence of distinct volatility regimes.

This evidence is visually summarized in Figure 1, which plots rolling-window second moments of the reduced-form residuals. The pronounced changes in variances and covariances across subperiods are consistent with the GI, GM, and GR+SR partition used in estimation and motivate the regime-based identification strategy.

4.3 Estimated impact matrices and exogeneity restrictions

Table 2 reports the estimated contemporaneous impact matrices for each volatility regime. The top panel presents the baseline specification, allowing the on-impact relations among macroeconomic uncertainty, industrial production growth, and financial uncertainty to vary across regimes. The bottom panel reports an alternative specification imposing additional restrictions designed to assess whether macroeconomic uncertainty can be treated as contemporaneously exogenous with respect to real-activity shocks. Likelihood-ratio statistics summarize the empirical support for these restrictions within the change-in-volatility identification framework.

Across regimes, the estimated impact matrices point to substantial heterogeneity in contemporaneous transmission mechanisms. In particular, the strength of the on-impact effect of macroeconomic uncertainty on real activity increases markedly in high-volatility environments, suggesting that contemporaneous interactions themselves are state dependent rather than merely scaled versions of a stable structure.

4.4 Impulse responses across volatility regimes

Figure 2 provides a compact overview of the impulse response functions across volatility regimes. Figures 3–5 report the regime-specific impulse responses with 90% bootstrap confidence bands. For each regime, blue lines correspond to one-month-ahead uncertainty shocks ($f = 1$), while red lines refer to one-year-ahead uncertainty shocks ($f = 12$).

A general result emerging from the analysis is that the transmission of uncertainty shocks is strongly regime dependent. Both the magnitude and the persistence of the responses vary substantially across volatility regimes, indicating that the relationship between uncertainty and real activity is not invariant over time.

In the Great Inflation regime, macroeconomic uncertainty shocks generate a clear contractionary effect on industrial production. The decline in real activity is statistically significant on impact and remains negative for several months before gradually reverting toward baseline. Financial uncertainty responds positively to macroeconomic uncertainty shocks, although the response is relatively contained and less persistent than in later regimes. Comparing short- and long-horizon uncertainty measures, one-year-ahead uncertainty shocks produce smoother but more persistent real effects, while one-month-ahead shocks generate sharper short-run responses. These patterns closely resemble those documented in earlier regime-based analyses and suggest that, during this period, uncertainty acted primarily as an exogenous driver of business-cycle fluctuations.

In the Great Moderation regime, responses are markedly weaker. Macroeconomic uncertainty shocks still induce a contraction in industrial production, but the magnitude is smaller and the effects dissipate more rapidly. Financial uncertainty reacts modestly, and feedback from real activity shocks to uncertainty remains limited. This attenuation of responses is consistent with the lower-volatility macroeconomic environment of the period and aligns with evidence

showing that uncertainty shocks tend to have muted real effects during tranquil times.

The strongest effects emerge in the Great Recession plus Slow Recovery regime. In this high-volatility environment, macroeconomic uncertainty shocks lead to large and persistent declines in industrial production, with output remaining below baseline for a prolonged horizon. Financial uncertainty responds sharply and displays substantial persistence, pointing to a powerful amplification mechanism. Differences between short- and long-horizon uncertainty shocks are particularly pronounced in this regime: long-horizon uncertainty shocks generate more persistent real effects, while short-horizon shocks produce sharper short-run contractions. These findings reinforce the view that uncertainty shocks are especially harmful during periods of elevated macro-financial stress.

4.5 Exogeneity or endogeneity?

Across all regimes, an important asymmetry emerges in the interaction between uncertainty and real activity. While uncertainty shocks consistently generate sizable and persistent responses in industrial production, especially in high-volatility regimes, shocks to real activity produce relatively limited and short-lived responses in both macroeconomic and financial uncertainty. This asymmetry supports the interpretation of uncertainty as a primary driver of business-cycle fluctuations rather than as a purely endogenous response to real activity shocks.

This result is particularly relevant in light of concerns raised in the literature regarding the endogenous nature of uncertainty measures. By allowing for contemporaneous feedback and exploiting changes in volatility for identification, the present framework directly addresses these concerns and provides evidence that uncertainty retains a structural role even once reverse causality is explicitly accounted for. At the same time, the pronounced state dependence of the responses helps reconcile these findings with evidence from alternative frameworks that emphasize stronger uncertainty effects during crisis periods.

Overall, the empirical evidence confirms that the macroeconomic effects of uncertainty shocks are highly state dependent and that macroeconomic uncertainty plays a causal role in driving real economic fluctuations, particularly during periods of elevated volatility.

5 Concluding Remarks

This paper reexamines the role of macroeconomic and financial uncertainty as drivers of U.S. business-cycle fluctuations using a nonrecursive SVAR framework identified through changes in volatility. Building on the approach of Angelini, Bacchiocchi, Caggiano and Fanelli (2019), the analysis extends the sample to June 2025 and evaluates whether the structural interpretation of uncertainty shocks remains robust in light of recent macroeconomic developments, including the COVID-19 pandemic and the subsequent period of elevated macro-financial volatility.

The empirical results provide strong support for the view that macroeconomic uncertainty

behaves as an exogenous source of business-cycle fluctuations. Across all volatility regimes, increases in macroeconomic uncertainty generate contractionary effects on real economic activity, with responses that are substantially stronger and more persistent during high-volatility periods. These findings confirm that the macroeconomic effects of uncertainty are highly state dependent and become particularly pronounced in environments characterized by heightened financial stress and elevated aggregate volatility.

Financial uncertainty, by contrast, plays a more indirect role. Its effects on real activity materialize mainly through its impact on macroeconomic uncertainty, with amplification mechanisms becoming more relevant from the mid-1980s onward. This pattern is consistent with the increasing importance of financial factors in shaping macroeconomic dynamics over the last four decades and helps reconcile the evidence presented here with existing contributions that emphasize the evolving role of financial markets.

Overall, extending the analysis to a longer and more recent sample does not overturn the main conclusions of the earlier literature but rather strengthens them. The evidence confirms that uncertainty shocks should be interpreted as regime-dependent structural disturbances rather than as purely endogenous responses to other shocks. From a methodological perspective, the results highlight the importance of adopting identification strategies that explicitly account for contemporaneous feedback, reverse causality, and structural instability. Ignoring these features risks obscuring the true causal role of uncertainty in macroeconomic fluctuations.

A natural extension of the present analysis would be to allow for an additional volatility regime explicitly associated with the COVID-19 crisis and the subsequent period of elevated inflation and uncertainty. While the current framework captures these developments within the last high-volatility regime, treating the pandemic and the post-pandemic inflationary episode as a distinct regime could provide further insights into whether the transmission of uncertainty shocks changed in nature, and not only in magnitude, during this exceptional period. Such an extension would also allow for a more precise characterization of uncertainty dynamics in the presence of unprecedented policy interventions and supply-side disruptions.

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Appendix: Tables and Figures

Table 1: Estimated reduced-form VAR covariance and correlation matrices

	Log-lik.	$\widehat{\Sigma}_\eta$ (covariances)		
		U_t^M	Y_t	U_t^F
Overall sample	3086.70	0.0001	-0.0008	0.0001
			0.7102	0.0021
				0.0008
		$\widehat{\rho}_\eta$ (correlations)		
		1.0000	-0.0795	0.3202
			1.0000	0.0910
				1.0000
GI	1170.50	0.0001	-0.0013	0.0000
			0.5874	0.0017
				0.0007
GM	1399.90	0.0001	-0.0006	0.0001
			0.2210	0.0004
				0.0008
GR+SR	770.01	0.0002	0.0020	0.0002
			1.0584	0.0051
				0.0007
Chow-type tests				
H_0: no breaks in VAR covariance matrix			$\mathrm{LR}_\Sigma = 507.4$	[0.000]

Notes. The table reports maximum-likelihood estimates of the covariance and correlation matrices of reduced-form VAR innovations for $X_t = (U_t^M, Y_t, U_t^F)'$ with four lags. GI denotes the Great Inflation, GM the Great Moderation, and GR+SR the Great Recession plus Slow Recovery. P-values are reported in brackets.

Table 2: Estimated structural parameters

GI: 1960:M8–1984:M3	GM: 1984:M4–2007:M12	GR+SR: 2008:M1–2025:M6

SVAR for $X_t := (U_t^M, Y_t, U_t^F)'$

$$\hat{B} := \begin{pmatrix} 0.0111 & -0.0010 & 0 \\ (0.0005) & (0.0008) & (0) \\ -0.0549 & 0.7645 & 0 \\ (0.0693) & (0.0324) & (0) \\ 0 & 0 & 0.0259 \\ (0) & (0) & (0.0011) \end{pmatrix} \quad \hat{B} + \hat{Q}_2 := \begin{pmatrix} 0.0045 & -0.0010 & 0.0071 \\ (0.0011) & (0.0008) & (0.0007) \\ -0.0240 & 0.4695 & 0 \\ (0.0446) & (0.0198) & (0) \\ -0.0177 & 0 & 0.0217 \\ (0.0025) & (0) & (0.0027) \end{pmatrix} \quad \hat{B} + \hat{Q}_2 + \hat{Q}_3 := \begin{pmatrix} 0.0045 & -0.0010 & 0.0150 \\ (0.0011) & (0.0008) & (0.0009) \\ -0.0379 & 1.0059 & 0.2126 \\ (0.0842) & (0.0498) & (0.0796) \\ -0.0177 & 0 & 0.0208 \\ (0.0025) & (0) & (0.0021) \end{pmatrix}$$

Model with "endogenous" macroeconomic uncertainty: $LR_T = 7.35\ \chi^2(df = 2)$ [0.025]

SVAR for $X_t := (U_t^M, Y_t, U_t^F)'$ (additional restrictions)

$$\hat{B} := \begin{pmatrix} 0.0112 & 0 & 0 \\ (0.0005) & (0) & (0) \\ -0.1203 & 0.7569 & 0 \\ (0.0455) & (0.0320) & (0) \\ 0 & 0 & 0.0259 \\ (0) & (0) & (0.0011) \end{pmatrix} \quad \hat{B} + \hat{Q}_2 := \begin{pmatrix} 0.0105 & 0 & 0.0028 \\ (0.0003) & (0) & (0.0006) \\ -0.0978 & 0.4641 & 0 \\ (0.0361) & (0.0194) & (0) \\ 0 & 0 & 0.0280 \\ (0) & (0) & (0.0012) \end{pmatrix} \quad \hat{B} + \hat{Q}_2 + \hat{Q}_3 := \begin{pmatrix} 0.0105 & 0 & 0.0085 \\ (0.0003) & (0) & (0.0008) \\ 0.0280 & 1.0111 & 0.1864 \\ (0.0558) & (0.0494) & (0.0704) \\ 0 & 0 & 0.0273 \\ (0) & (0) & (0.0014) \end{pmatrix}$$

Model with "exogenous" macroeconomic uncertainty: $LR_T = 8.03\ \chi^2(df = 4)$ [0.091]

Notes. Estimated structural parameters from a nonrecursive SVAR identified through changes in volatility. Y_t denotes industrial production growth. Standard errors based on the Hessian are reported in parentheses. GI, GM, and GR+SR denote the Great Inflation, the Great Moderation, and the Great Recession plus Slow Recovery, respectively.

Figure 1: Second moments of reduced-form VAR innovations. The figure plots recursive and rolling-window estimates of variances and covariances of the reduced-form residuals for (U_t^M, Y_t, U_t^F). Vertical markers indicate the break dates separating GI, GM, and GR+SR.

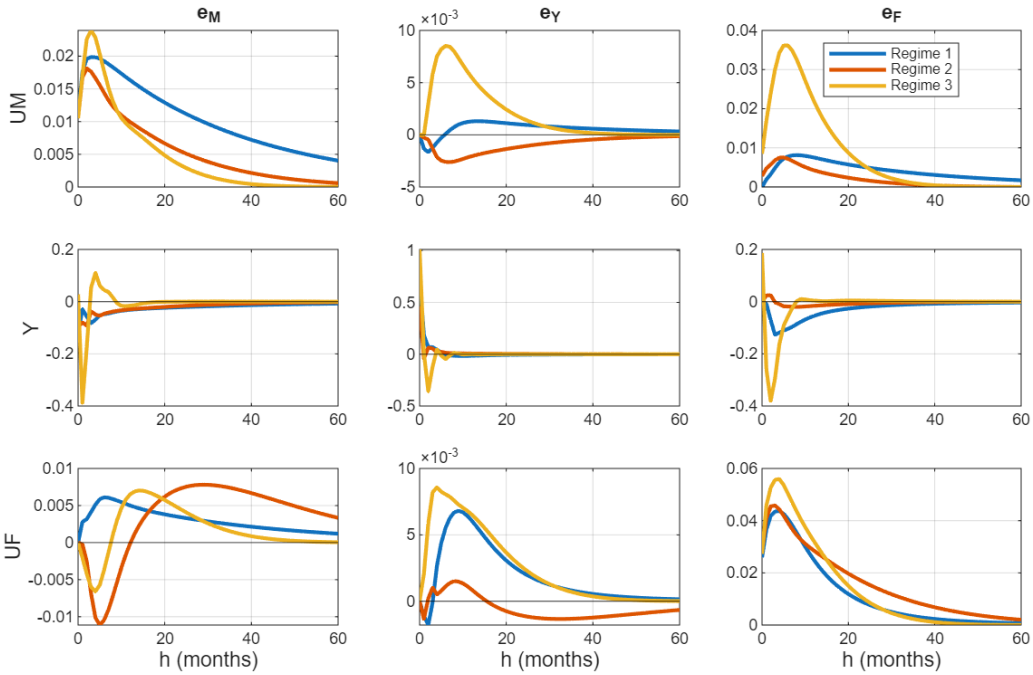

Figure 2: Structural impulse responses across volatility regimes. Each panel reports the response of (U_t^M, Y_t, U_t^F) to one-standard-deviation structural shocks. Responses are shown up to $h = 60$ months.

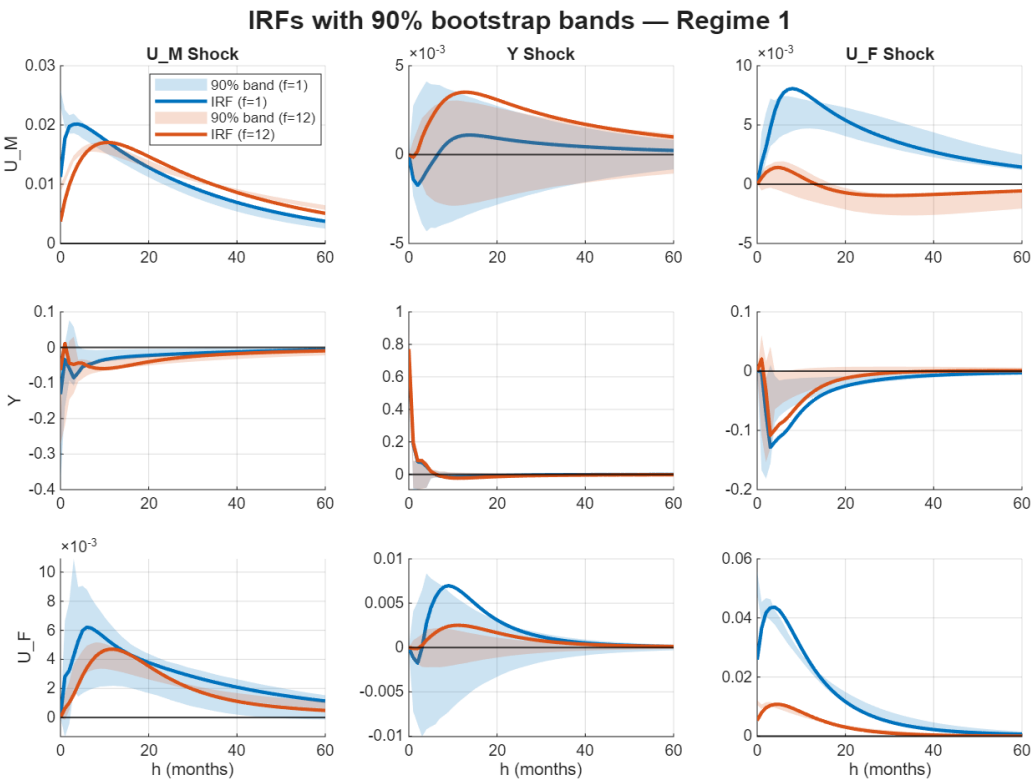

Figure 3: IRFs with 90% bootstrap bands — GI. Blue lines refer to shocks identified using one-month-ahead uncertainty ($f = 1$), while red lines refer to one-year-ahead uncertainty ($f = 12$). Shaded areas denote 90% bootstrap confidence bands.

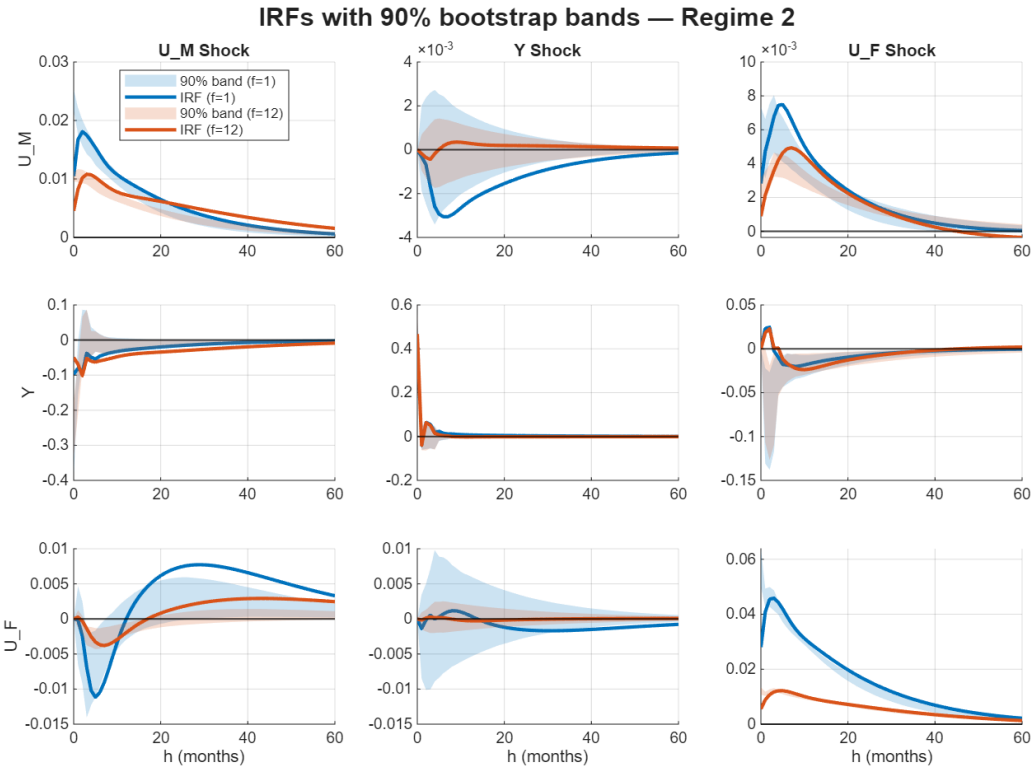

Figure 4: IRFs with 90% bootstrap bands — GM. Blue lines: $f = 1$. Red lines: $f = 12$. Shaded areas denote 90% bootstrap confidence bands.

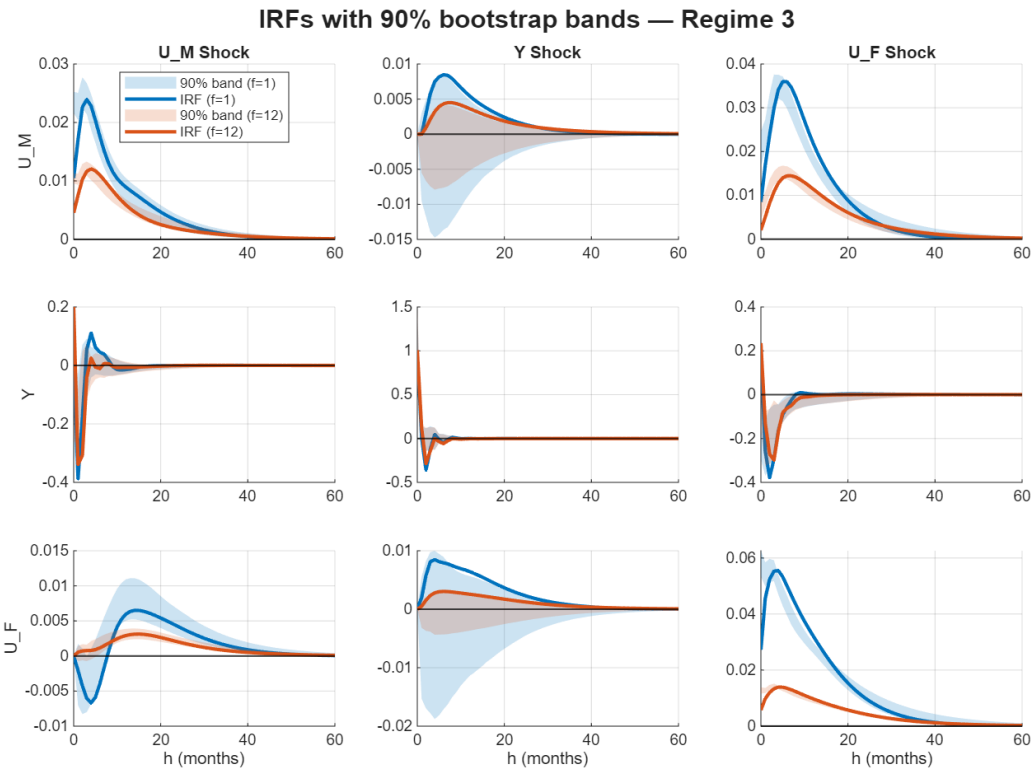

Figure 5: IRFs with 90% bootstrap bands — GR+SR. Blue lines: $f = 1$. Red lines: $f = 12$. Shaded areas denote 90% bootstrap confidence bands.